- In the following code, what is the printout for list2?

```
class  Test {
  public static void main(String[] args) {
     int[] list1 = {3, 2, 1};
     int[] list2 = {1, 2, 3};
     list2 = list1;
     list1[0]   = list1[0] * list2[0];
     list1[1]  = list1[1] * list2[1];
     list1[2]  = list1[2] * list2[2];
     for(int i = list2.length – 1; i >= 0; i--)
            System.out.print(list2[i] + " ");
```

- If you declare an array double[] list = {3.4, 2.0, 3.5, 5.5}, list[1] is ________.

}

- Write the following method that returns true if the list is already sorted in increasing order.

public static boolean isSorted(int[] list)

<Output>
Enter list: 8 10 1 5 16 61 9 11 1 The list is not sorted
<End Output>

<Output>
Enter list: 10 1 1 3 4 4 5 7 9 11 21 The list is already sorted

- Write a method that computes the number of letters in a String:

The method signature is as follows:

```
Public int count(String s)
{


}
```

- Write a method called *count* that counts number of occurrences of all odd elements in an array of integer values. For example, if:

List

5	10	15	5	30	43	5	60	70	80

Then count(list) will return 5

```
public static int count(int[] list)
{


}
```

- _________ describes the state of an object.
  a. data fields
  b. methods
  c. constructors
  d. none of the above

- An attribute that is shared by all objects of the class is coded using ________.

  a. an instance variable
  b. a static variable
  c. an instance method
  d. a static method

Given the following class:

```
  public class A {
    private int a;
    private double b;
    private boolean c;
    private Date d;
}

public A() {

}
```

- What is the default value of each of the data fields?

- Write an overloaded constructor to initialize the data fields with the given arguments.

- If a class named Student has a constructor Student(String name) defined explicitly, the following constructor is implicitly provided.

a. public Student()
b. protected Student()
c. private Student()
d. Student()
e. None

- Suppose the xMethod() is invoked from a main method in a class as follows, xMethod() is _________ in the class.

```
public static void main(String[] args) {

  xMethod();
 }
```

a. a static method
b. an instance method

- What would be the result of attempting to compile and run the following code?

```
public class Test {
  static int x;

  public static void main(String[] args){
        System.out.println("Value is " + x);
  } }
```

a. The output "Value is 0" is printed.
b. An "illegal array declaration syntax" compiler error occurs.
c. A "possible reference before assignment" compiler error occurs.
d. A runtime error occurs because x is not initialized.

- How can you get the word "abc" in the main method from the following call?

```
java Test "+"  3  "abc"  2
```

a. args[0]
b. args[1]
c. args[2]
d. args[3]

- Write an array called “list”, write a method to return the number of odd elements in the list.

- Tanks containing some fluid are described by the following class

```
public  class Tank
{
        private int capacity;    // capacity of the Tank
        private int level;       // level of the fluid in the Tank (must be <= capacity

        // Create an empty Tank with capacity of c
        public  Tank(int c)
        {
                capacity = c;
                level = 0;
        }

        // Fill up this tank to capacity
        public  void  fill()
        {
                level = capacity;
        }

        // Empty out this Tank
        public  void empty()
        {
                Level = 0;
        }

        // Get the capacity of this Tank
        public  int getCapacity()
        {
                return capacity;
        }

        // get the level of this Tank
        public int getLevel()
        {
                return level;
        }

// Pour fluid from other Tank into this tank
        public  void pourFrom(Tank other)
        {

        }
}
```

*Implement the method pourFrom(Tank other), which pours liquid from Tank other into Tank this. For example, the call t1.pourFrom(t2) should pour all of the liquid from t2 into t1 unless this would cause Tank t1 to overflow. In that case, t1.pourFrom(t2) should fill up Tank t1 and leave the remainder of the liquid in Tank t2.

```
 public void pourFrom(Tank other)
{


}
```

Given two tanks with capacity of 7 and 4, write a complete java program TestTank.java to produce 5 units of fluid in larger tank. The algorithm is:

Create the smaller tank
Create the larger tank
Fill the smaller tank
Pour it into the larger tank
Fill the smaller tank again
Pour into larger one till filled
Empty the larger tank
Pour the smaller tank into larger tank
Fill the smaller tank
Pour smaller tank into larger tank

Review

Money.java, Length.java
Algorithms

isPrime(int n)
selectionSort(int[] list)
binarySearch(int[] list, int key)
linearSerach(int[] list, int key)
isPalindrome(String s)
arrays (find sum, max, avg, reverse)
Two-dimensional array(find max, sum all element, sum by row, sum by column)

generate a list of lowerCase/upperCase letters